

Armaan Amatyia

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EDUCATION

University of Houston

Bachelor of Science in Computer Science and Mathematics, GPA: 3.5

Houston, TX

Aug. 2024 – Dec. 2026

EXPERIENCE

Autonomize AI

Austin, TX

Machine Learning Engineer Intern

May 2026 – Aug. 2026

- Quantized Qwen2.5-VL-32B with 4-bit AWQ and post-trained it with RLHF for multimodal OCR at 95% of baseline accuracy, then served it on a 16GB Azure T4 with vLLM, request batching and CPU offloading, cutting VRAM 4x
- Built visual token compression for production OCR workflows on Azure and GCP, pruning 50%+ redundant image tokens to cut inference latency 35% and KV-cache memory 40% and raise throughput 1.5x at under 2% accuracy loss
- Shipped a RAG agent over 100+ policy docs saving 20+ hours/week; 90%+ accuracy at 2s per question, batched

University of Houston

Houston, TX

Undergraduate Researcher

Jan. 2026 – Present

- Cut multimodal LLM inference latency 40% via a novel token compression method that reduces visual tokens 60%, shortening attention context and KV-cache footprint at 98% of baseline accuracy on Qwen2.5/3-Omni
- Deployed compressed models to NVIDIA Jetson Orin for on-device inference after multi-GPU distributed training (PyTorch DDP), cutting VRAM 45% and sustaining 2x throughput (tokens/s); won Best Research Award

Artinafti

Austin, TX

AI Engineer Intern

Jan. 2026 – May 2026

- Improved visual fidelity 25% by fine-tuning production diffusion upscaling and generation models with LoRA adapters and detail-preserving objectives for text and faces, adding photographic color-tone control
- Deployed model-serving APIs (FastAPI, NestJS, Redis) serving live users; request batching cut latency 30%

AutoHDR

Austin, TX (Remote)

Machine Learning Engineer Intern

Jan. 2026 – Apr. 2026

- Cut per-image turnaround from 5 minutes to under 15 seconds across 10,000+ listing photos monthly by building high-throughput diffusion pipelines (PyTorch) with queue-backed batch GPU inference
- Cut redundant processing 25% with contrastive CLIP dedup batched into the production GPU pipeline
- Produced 1,000+ property tour videos monthly with a depth-aware image-to-video generation pipeline

FuseMachines

New York, NY

Machine Learning Engineer Intern

May 2024 – Aug. 2024

- Achieved 98% precision on real-world attendance tracking by fine-tuning a FaceNet model
- Served real-time inference to a production mobile app (FastAPI, Docker) with async batched requests

PROJECTS

Mixed-Precision GEMM Kernels in Triton | [Link](#) | Triton, CUDA, PyTorch, Nsight Compute

- Matched vendor BLAS with a mixed-precision FP16 GEMM kernel in Triton, hitting 104% of cuBLAS on Blackwell and 98% on Ampere at N=8192 across 10 independent runs per point
- Traced the win through a V1–V5 ablation across 3 GPUs and 2 vendors: 128x256 tiling alone regressed throughput up to 8x on Ampere until 3-stage software pipelining, with a 192-config autotune sweep closing the gap and Nsight Compute profiling (occupancy, memory bandwidth, kernel launch overhead) isolating the bottleneck at each step

Real-Time Distributed Keyword Spotting | [Link](#) | PyTorch, Wav2Vec2, CUDA, VxWorks, UDP

- Held the CUDA Wav2Vec2 node to 8 ms average and 12.5 ms p99 over 484 inferences against a 150 ms hard deadline for emergency voice commands, despite an RMS bound of U above 1 predicting the task set unschedulable
- Found FP16 a net loss for single-sample Transformer inference: vs. FP32 it raised average latency 7% and e2e p99 1.5x (12.5 to 18.6 ms) across 480+ inferences, overturning the expected speedup at batch size 1

Domain QA System for Scientific PDFs | [Link](#) | HuggingFace, PEFT, BitsAndBytes, FAISS, LangChain

- Post-trained Mistral-7B with QLoRA (4-bit NF4 quantization) on 234 QA pairs; built hybrid RAG over a 75-paper FAISS corpus; evaluated across BLEU, BERTScore, and GPT-4o-as-judge to catch metric-specific blind spots before shipping quantized GenAI inference on HuggingFace Spaces

TECHNICAL SKILLS

Languages: Python, C++, CUDA, Triton, TypeScript, JavaScript, Go, Rust, SQL, Bash

Frameworks: PyTorch, vLLM, TensorRT, HuggingFace, LangChain, FastAPI, React, Next.js, Node.js, NestJS

Cloud: AWS, Azure, GCP, Docker, Kubernetes, PostgreSQL, Redis, FAISS, inference optimization, request batching

AI/ML: LLMs & model architectures, transformers & attention, RL (PPO/GRPO/RLHF), post-training & alignment, ML pipelines, Azure model deployment, RAG, quantization (4-bit AWQ/QLoRA), kernel optimization, numerics validation